

LESSON 2.5

APPROXIMATING IRRATIONAL NUMBERS



LESSON INTRODUCTION

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers represented in various forms.

LEARNING TARGETS

- I can identify whether a number is rational or irrational.
- I can approximate the values of irrational numbers.
- I can plot irrational numbers on a number line.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

WORKING TOWARD

912.NSO1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

ESSENTIAL QUESTIONS

- How do you identify whether a number is rational or irrational?
- How do you approximate the value of irrational numbers?
- How do you plot irrational numbers on a number line?

LESSON NARRATIVE

Students compare rational numbers and irrational numbers and identify each using definitions. They approximate nonperfect square and cube roots and plot the approximated values on a number line. This work will prepare them to compare, order, and plot rational and irrational numbers on number lines in the next two lessons.

COMPONENT SUMMARY

2.5.1 WARM-UP (~5 MIN.)

Identify numbers as rational or irrational

2.5.3 GUIDED INSTRUCTION (5-10 MIN.)

Define and compare rational and irrational numbers

2.5.5 GUIDED INSTRUCTION (10 MIN.)

Approximate irrational numbers and plot them on a number line

2.5.7 WRAP-UP (~5 MIN.)

Reflect on learning during today's lesson

2.5.2 COLLABORATION (5 MIN.)

Identify and correct an error in approximating the value of a square root and plotting it on a number line

2.5.4 COLLABORATION (5-10 MIN.)

Determine whether numbers are rational or irrational

2.5.6 GUIDED PRACTICE (10 MIN.)

Practice approximating irrational numbers and plotting them on a number line

RESOURCES

GLOSSARY

Key Terms:

- Rational number
- Irrational number

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Students will need access to a calculator for 2.5.3 Guided Instruction, but then calculators should not be used for the remainder of this lesson or for the subsequent lessons.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle approximating irrational square and cube roots to the nearest tenth.
- Students may mistakenly think that pi equals 3.14 or $\frac{22}{7}$ rather than recognizing these are only approximations.
- Students may mistakenly identify $\frac{\pi}{3}$ as rational because it is written as a ratio.

THINGS TO CONSIDER

- Remind students that when approximating the value of irrational numbers, first consider which whole numbers the nonperfect square or cube root falls in between. Then consider which whole number it is closer to. Finally, test a few decimals by squaring or cubing them to determine which is the best approximation.
- If time is an issue, then consider assigning 2.5.6 Guided Practice for homework.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 2.5.2 Collaboration, students work together as they solidify their thinking around how to approximate the square root of a nonperfect square number and plot it on a number line. Prompt students to take turns as they discuss the questions. This routine can also be used in 2.5.4 Collaboration.

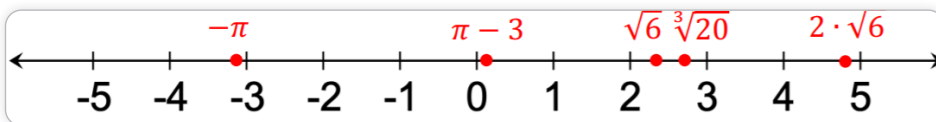
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students who exhibit mathematical fluency can “adapt procedures to apply them to a new context.” Students will be identifying irrational numbers and approximating square and cube roots and plotting them on number lines by using what they know about rational numbers and approximating square and cube roots. They will continue to develop fluency with this skill.

DIFFERENTIATE INSTRUCTION

Consider creating a classroom number line where students can plot the square and cube roots during the debrief of 2.5.2 Collaboration, 2.5.5 Guided Instruction, and 2.5.6 Guided Practice. Physically plotting the points on a classroom number line and explaining their placement of the point provides kinesthetic, visual, and audio entry points.



Additional differentiation opportunities are denoted throughout the lesson guide.

2.5.1 WARM-UP: RATIONAL VS. IRRATIONAL

TEACHER MOVES

This Warm-Up requires students to recall the definitions of rational and irrational numbers. Prompt students to use strategic trial and error, keeping track of the substitutions that they tried and that did not work.

QUESTIONING

Students worked with rational numbers in previous courses and discovered that square roots of nonperfect squares are irrational in the previous lesson. If students struggle, do not use this time to teach the concepts. These definitions will be clarified in 2.5.3 Guided Instruction. Consider allowing students to return to this puzzle to check their work and make any needed revisions after completing 2.5.3 Guided Instruction.



2.5.1 Warm-Up: Rational vs. Irrational

- Complete the table by filling in values in each box with a digit, 1 through 8, such that the value of the expression matches the given description. Each digit can only be used once.

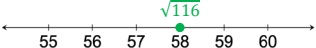
Answers will vary.

Expression	Description
$\sqrt{\boxed{4}}$	This is a rational number.
$\sqrt{\boxed{8}}$	This is an irrational number.
$\frac{\boxed{6}}{\boxed{2}}$	This is a rational number and an integer.
$\frac{\boxed{3}}{\boxed{5}}$	This is a rational number with a terminating decimal expansion.
$\frac{\boxed{1}}{\boxed{7}}$	This is a rational number with a repeating decimal expansion.

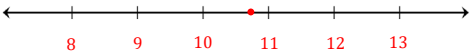


2.5.2 Collaboration: Error Analysis

Robin is learning about square roots. A question and her response are shown in the table. Since this is new for Robin, there are some mistakes in her response.

Question	Robin's Response
Where would $\sqrt{116}$ be located on a number line?	Taking the square root is the inverse of squaring a number. Squaring a number is the same as raising it to the second power. To find the value of a square root, you divide the radicand by 2. The number line shows the location of $\sqrt{116}$ because $116 \div 2 = 58$. 

1. Discuss Robin's response with your partner. Then, cross out any incorrect statements in her response.
2. Work with your partner to write a correct explanation of how to locate $\sqrt{116}$ on a number line.
Answers will vary. $\sqrt{116}$ is not a perfect square but the $\sqrt{100} = 10$ and the $\sqrt{121} = 11$. 116 is closer to 121 so the $\sqrt{116}$ is located between 10 and 11 on the number line, but closer to 11. $10^2 = 100$; $11^2 = 121$ so $\sqrt{116}$ should be > 10 but < 11 and closer to 11 on the number line.
3. Use the number line to plot and label a point to better approximate the location of $\sqrt{116}$. Be sure to write an appropriate scale using the tick marks.



2.5.2 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

Take Turns

Students work together as they solidify their thinking around how to approximate the square root of a nonperfect square number and plot it on a number line. Prompt students to take turns as they discuss the questions.

ENRICHMENT

- Consider asking any of the following questions as needed.
- What steps did you take to approximate the square root?
 - Which whole numbers is the square root in between?
 - Which whole number is the square root closer to?
 - How many tick marks are on the provided number line? Which values should you include on the number line so you can represent the approximation of $\sqrt{116}$?

DIFFERENTIATE INSTRUCTION

Consider providing the scale for the number line so students can focus on determining how to approximate $\sqrt{116}$ and represent the approximation on the number line. Students may also benefit from an anchor chart with a list of perfect squares from 1 to 15.

2.5.3 GUIDED INSTRUCTION: RATIONAL AND IRRATIONAL NUMBERS

TEACHER MOVES

In Lesson 4, students discovered that the square and cube roots of nonperfect squares and cubes are irrational numbers. In this lesson, students apply this knowledge to classify numbers as rational or irrational. Ask students to share their reasoning as they determine whether numbers are rational or irrational throughout the lesson, referring back to the definitions.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

As students discuss what they recall about pi, listen for misconceptions (i.e. claiming that pi equals 3.14 or $\frac{22}{7}$, because these are approximations) and address in question 1b to ensure students recognize pi as an irrational number.

TEACHER MOVES

Completion of these statements should be led by the teacher. The teacher may decide to call on students to share their thinking at various points based on what they heard when students discussed question 1a.



2.5.3 Guided Instruction: Rational and Irrational Numbers

The real number system includes both **rational** and **irrational numbers**.



A **rational number** is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.



An **irrational number** is a real number that cannot be expressed as a ratio of two integers.

When expressed as decimals, irrational numbers do not terminate or repeat. Therefore, any decimal expansion is only an approximate value.

1. The relationship between the circumference and diameter of any circle is represented by pi (π).
 - a. Discuss with your partner what you remember about the value of π .
Answers will vary. Pi is the ratio of the circumference to the diameter of a circle. The circumference divided by the diameter results in pi every time. You can multiply a circle's diameter by pi to get its circumference. Pi is 3.14159... it goes on forever and never repeats, which means it is an irrational number. Sometimes we use 3.14 or $\frac{22}{7}$ to approximate the value of pi when solving problems.

- b. Complete the statements.

The ratio of the circumference of a circle to its diameter is expressed as π . The values 3.14 and $\frac{22}{7}$

are ☐ equivalent to, or the exact value of π .
☒ used to approximate the value of π .

The exact value of π ☐ can ☒ cannot be expressed as a

fraction of two integers or a decimal. Therefore, π is

☐ a rational ☒ an irrational number.

2. Enter each of the following values into a calculator.

$\sqrt{25}$

$\sqrt{18}$

$\sqrt[3]{27}$

$\sqrt[3]{30}$

- a. Circle the values that can be expressed as a ratio of two integers.
- b. Discuss with your partner how you would describe the values that are not circled.
Answers will vary. The square and cubed roots of the values not circled are not perfect squares or perfect cubes so they are irrational numbers. The square and cubed roots of the values not circled are non-perfect therefore they are irrational numbers.
- c. Complete the statements.

Square roots of perfect squares and cube roots of perfect cubes ☐ can ☐ cannot be expressed as a ratio of two integers. Therefore, they are ☐ rational. ☐ irrational.

Square roots of non-perfect squares and cube roots of non-perfect cubes ☐ can ☐ cannot be expressed as a ratio of two integers. Therefore, they are

☐ rational.
☐ irrational.

2.5.3 GUIDED INSTRUCTION: RATIONAL AND IRRATIONAL NUMBERS (CONTINUED)

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address when completing the statements in question 2c.

TEACHER MOVES

If time permits, allow students to go back to the 2.5.1 Warm-Up to revise their work. Invite students to share their reasoning for any changes they made. Consider using this as another opportunity for students to display their understanding.

2.5.4 COLLABORATION: RATIONAL OR IRRATIONAL?

TEACHER MOVES

Questions 2 and 3 are intended to draw out specific potential misconceptions. For question 2, students may identify $\pi/3$ as rational because it is written as a ratio. Discuss how for it to be rational, it must be a ratio of integers. For question 3, students may not recognize the notation for a repeating decimal or recall from 7th grade that repeating decimals are rational and can be represented as a ratio of integers.

QUESTIONING

Consider asking students who finish early to create a few additional expressions for each column. Prompt students to include a justification for each of the expressions they add to the columns.

DIFFERENTIATE INSTRUCTION

Some students may struggle composing explanations in questions 2 and 3. Consider providing students with sentence stems or answer choices.



2.5.4 Collaboration: Rational or Irrational?

Work with your partner to complete the following.

Several numbers are given:

$$-\frac{4}{5} \quad \frac{\pi}{3} \quad 5.124784 \dots \quad 4.\overline{35} \quad 3.275 \quad -\sqrt{40} \quad \sqrt[3]{-27}$$

- Write each expression in the appropriate column of the table based on the definitions of rational and irrational numbers.

Rational	Irrational
$-\frac{4}{5}$	$\frac{\pi}{3}$
$4.\overline{35}$	$5.124784 \dots$
3.275	$-\sqrt{40}$
$\sqrt[3]{-27}$	

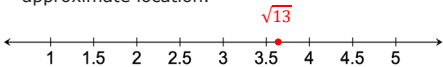
- Explain how you determined whether $\frac{\pi}{3}$ is rational or irrational.
Answers will vary. I determined it is irrational because π is irrational. Dividing π by 3 would give you another irrational result. $\frac{\pi}{3}$ is irrational because rational numbers must be a ratio of integers and π is not an integer.
- Explain how you determined whether $4.\overline{35}$ is rational or irrational.
Answers will vary. $4.\overline{35}$ is rational because the bar over the decimal part means that those numbers repeat. Repeating decimals are rational numbers. In decimal form, rational numbers either terminate or repeat.



2.5.5 Guided Instruction: Approximating Irrational Numbers and Locating Them on a Number Line

The values of irrational numbers and expressions involving irrational numbers can be approximated.

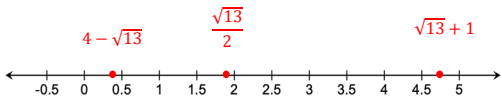
1. Consider the expression $\sqrt{13}$.
- a. Between which two whole numbers does $\sqrt{13}$ lie on a number line?
3 and 4
 - b. Which whole number is $\sqrt{13}$ is closer to?
4
 - c. Faith thinks the value of $\sqrt{13}$ will be close to 3.6 or 3.7. Discuss with your partner why Faith may have made these estimates.
Answers will vary. $3^2 = 9$ and $4^2 = 16$. 13 is a little more than halfway between the perfect squares 9 and 16. Faith estimates 3.6 or 3.7 because that is a little more than halfway between 3 and 4 on the number line.
 - d. Check Faith's estimates to determine if either estimate is close to the value of $\sqrt{13}$.
 $3.7 \cdot 3.7 = 13.69$ whereas $3.6 \cdot 3.6 = 12.96$ so $\sqrt{13}$ is closer to 3.6.
 - e. Based on your work in Part D, complete the statement.
The approximate value of $\sqrt{13}$, to the nearest tenth, is *3.6*.
 - f. Plot and label $\sqrt{13}$ on the number line to indicate its approximate location.



2. Use the approximation of $\sqrt{13}$ from the previous question to complete the following.
- a. Approximate the value of each expression.

Expression	Approximate Value
$\sqrt{13} + 1$	<i>4.6</i>
$\frac{\sqrt{13}}{2}$	<i>1.8</i>
$4 - \sqrt{13}$	<i>0.4</i>

- b. Plot and label the approximate location of each expression from part a on the number line.



To approximate the value of expressions involving irrational numbers, first approximate the value of the irrational number. Then, use that approximation to perform any operations indicated.

2.5.5 GUIDED INSTRUCTION:
APPROXIMATING IRRATIONAL NUMBERS
AND LOCATING THEM ON A NUMBER LINE

TEACHER MOVES

Model squaring 3.6 and 3.7 to determine whether the decimal approximation for $\sqrt{13}$ is closer to 3.6 or 3.7. Explain that $(3.6)^2 = 12.96$ and $(3.7)^2 = 13.69$. Therefore, 3.6 is a closer approximation.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

ENRICHMENT

- While modeling how to approximate the value of each expression, consider using a Think Aloud, asking and answering the following:
- How did you approximate each expression?
 - In each expression _____, which operation is performed first?

TEACHER MOVES

Use a Think Aloud to model how to plot and label each expression on the number line.

2.5.6 GUIDED PRACTICE: APPROXIMATING IRRATIONAL NUMBERS AND LOCATING THEM ON A NUMBER LINE

DIFFERENTIATE INSTRUCTION

Consider creating a classroom number line during the debrief of this activity. Physically plotting the points on a classroom number line and explaining their placement of the points provides kinesthetic, visual, and audio entry points.

ENRICHMENT

If needed, consider asking the following scaffolding questions.

- How did you determine which whole numbers the approximation falls in between?
- How did you determine the decimal approximation of each expression?
- How can you check the accuracy of your approximations?

TEACHER MOVES

If time is an issue, then consider assigning this Guided Practice for homework.

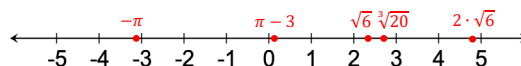


2.5.6 Guided Practice: Approximating Irrational Numbers and Locating Them on a Number Line

1. Approximate the value of each expression.

Expression	Approximate Value
$-\pi$	-3.1
$\sqrt{6}$	2.4
$\sqrt[3]{20}$	2.7
$2 \cdot \sqrt{6}$	4.8
$\pi - 3$	0.1

2. Plot and label the approximate location of each expression from the previous question on the number line.





2.5.7 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were...	Something I wondered during the lesson was...	Something I learned during lesson was...
<i>Answers will vary.</i> • squares • square roots • cubes • cube roots • rational numbers • irrational numbers • perfect • non-perfect	<i>Answers will vary.</i> • Why are repeating decimals considered rational? • If there are other types of roots besides square roots and cubed roots?	<i>Answers will vary.</i> • How to approximate the value of expressions involving irrational numbers. • How to perform operations with approximate values.

2.5.7 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.